2 Axis Gimbal

Overview

Designed and developed a two-axis robotic mechanism capable of positioning an end effector in a 2D plane using two motors driving interconnected arms in a parallelogram configuration. Dedicated two weeks to designing, testing, and performing Finite Element Analysis (FEA) on the cycloidal drives used to power the mechanism, evaluating their structural performance and refining the designs. Derived the inverse kinematics equations by hand to calculate the required motor angles for a given (x,y) end-effector coordinate, establishing the mathematical foundation for precise planar positioning.

Design

Cycloidal Gearbox Design

I developed an initial cycloidal drive prototype for a NEMA 17 motor with a 15:1 reduction ratio and a footprint matching the motor's horizontal dimensions. The output was positioned at the top of the gearbox and featured a four-bolt mounting pattern for easy attachment integration. This first iteration focused on validating the mechanism's functionality, testing basic motor control code, and evaluating its accuracy and repeatability.

One major issue with the initial design was that the cycloidal discs could wobble within the gearbox despite being constrained coaxially with the shaft. I initially addressed this by adding M3 washers to the output pins to maintain separation between the discs. However, the stationary washers introduced additional friction as the discs moved against them. In the second iteration, I incorporated grooves into the discs to house bearing balls that moved with the discs, reducing wobble while maintaining smooth motion. The final design achieved a 26:1 reduction ratio within a 70 mm diameter footprint. Because the discs were constrained to a fixed angular orientation and only oscillated vertically, the output was relocated to the outer drum, where the cycloidal lobes were positioned.

Two-Axis Robotic Arm

I used two of the finalized cycloidal drives as actuators for a custom two-axis robotic arm. I stacked the drives vertically and attached a custom two-link arm to each actuator through the outer drum outputs. I then connected the two arms to form a parallelogram linkage, enabling coordinated planar motion of the end effector.

Programming

Don’t currently have access to files as this is a new laptop and i forgot to upload to git oops.